

# Logic

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# Chapter Summary

- 1 Introduction
- 2 Propositional logic
- 3 Predicate logic
- 4 Proof techniques

# What is Mathematics, really?

- It's not just about numbers!
- Mathematics is much more than that:
- Mathematics is, most generally, the study of **any and all** absolutely certain truths about **any and all** perfectly well-defined concepts.
- These concepts can be about numbers, symbols, objects, images, sounds, anything!
- It is a way to interpret the world around you.

# So, what's this class about?

- What are “**discrete structures**” anyway?
- “**Discrete**” - Composed of distinct, separable parts. (Opposite of continuous.)
- **Example:** Natural numbers are discrete, i.e., 2, 3, 4, ...  
Real numbers are continuous, i.e., numbers between 0 and 1
- “**Structures**” - Objects built up from simpler objects according to some definite pattern.
- “**Discrete Mathematics**” - The study of discrete, mathematical (i.e., well-defined conceptual) objects and structures.
- It is not a branch of mathematics. It is rather a description of the set of branches that have a common property - that they are discrete, not continuous

# Why Discrete Mathematics?

Discrete Mathematics is essential to Computer Science!

- Various aspects of discrete mathematics form the foundation for:
  - modelling computing structures
  - designing programs and algorithms
  - reasoning about programs and algorithms
  - solving real-world problems using the computer
  - in particular, solving problems in information technology, i.e. problems involving computers, communication, information.
- Aspects of discrete mathematics: (David J. Hunter, Essential of Discrete Mathematics)
  - logical thinking
  - relational thinking
  - recursive thinking
  - quantitative thinking (counting)
  - analytical thinking

**Question:** How do these 5 aspects appear in the activities listed above?

# Logical Thinking

- formal logic: (symbolic manipulation of notation; logical not-thinking :-)))) )
- propositional logic: propositional calculus allows us to make logical deductions formally
- predicate logic: make a proposition to depend on a variable and we get a predicate; here the logical deductions include quantifiers (for all, there exists) in front of the predicates)
- methods of proof: direct, by contraposition, by contradiction use what you learned in formal/symbolic logic, to guide your reasoning on mathematical proofs (written in paragraph form)
- logic in programming imperative programming: conditional statements (if-then-else, do-while) logic programming languages (e.g. prolog): uses the rules of predicate logic
- logic in circuits

# Relational Thinking

- It deals with the following type of structures:
  - sets
  - functions
  - relations
  - partial orderings
  - graph theory

Question 1: what are each of these structures?

Question 2: give examples of situations where they can be applied in computer science.

- Databases: table=relation; record= $n$ -ary tuple
- Dependency of task executions (partial ordering); topological sorting: order tasks respecting dependencies.
- Graphs: networks (communication, roads, social), conflicts (timetabling, coloring maps), hierarquies (rooted trees), diagrams (binary relations).

# Recursive Thinking

- Recurrence relations  
Recursively defined sequences of numbers. e.g. Fibonacci sequence.
- Recursive definitions  
e.g. binary trees, recursive geometry/fractals
- Proofs by induction  
Prove that  $P(n)$  is true for all  $n \geq 0$ : basis:  $P(0)$  is true + induction step  $P(n) \Rightarrow P(n + 1)$
- Recursive data structures  
e.g. binary search trees
- Recursive algorithms  
e.g. binary search, mergesort, solving towers of Hanoi.



# Quantitative Thinking

- counting,
- combinations, permutations, arrangements,
- the pigeonhole principle,
- discrete probability,
- counting operations in algorithms,
- estimating growth of functions, big-Oh notation.

# Analytical Thinking

- Apply previous tools to analyze problems of interest such as:
  - Writing programs that are correct.  
software engineering tools: testing versus verification.
    - loop invariants,
    - program correctness and verification.
  - Writing algorithms that are efficient.
    - algorithm complexity,
    - analysis of algorithms.

Question: How previous tools can be applied in each of the above areas?

- This question will be answered more fully by the studies in this course.

# Logical Thinking

# Propositional Logic

**Sentences:** Classified as declarative, exclamatory, interrogative, or imperative.

**Proposition:** Basic building blocks of logic.  
a declarative sentence (a sentence that declares a fact) that is either *true* or *false*, but not both.

## Assumptions about propositions:

**Assumption 1:** The Law of the Excluded Middle. For every proposition  $p$ , either  $p$  is true or  $p$  is false.

**Assumption 2:** The Law of Contradiction. For every proposition  $p$  it is not the case that  $p$  is both true and false.

# Propositional Logic

All the following declarative sentences are propositions.

- ① New Delhi is the capital of India.
- ② Toronto is the capital of Russia.
- ③  $1 + 1 = 2$ .
- ④  $2 + 2 = 3$ .

Propositions 1 and 3 are *true*, whereas 2 and 4 are *false*.

- ① What time is it? (**NO!**, not declarative sentence)
- ② Read this carefully. (**NO!**, not declarative sentence)
- ③  $x + 1 = 2$ . (**NO!**, neither *true* nor *false*)
- ④  $x + y = z$ . (**NO!**, neither *true* nor *false*)

**Note:** Sentences 3 and 4 can be turned into a proposition if we assign values to the variables.

# Propositional Logic

- letters  $p, q, r, s, \dots \Rightarrow$  propositional/statement variables
- The truth value of a proposition is *true*, denoted by  $T$ , if it is a true proposition,
- The truth value of a proposition is *false*, denoted by  $F$ , if it is a false proposition.
- The area of logic that deals with propositions is called **propositional calculus** or **propositional logic**.
- It was first developed systematically by the Greek philosopher Aristotle more than 2300 years ago.

# Propositional Logic

- Methods for producing new propositions from those that we already have.
- These methods were discussed by the English mathematician George Boole in 1854 in his book “The Laws of Thought”.
- Many mathematical statements are constructed by combining one or more propositions. New propositions, called **compound propositions**, are formed from existing propositions using logical operators.

# Propositional Logic: Connectives

Five Basic Connectives.

Three basic

- 1 not (modify a proposition) ( $\neg$ )
- 2 and ( $\wedge$ )
- 3 or ( $\vee$ )

Two conditional

- 1 if ... then ( $\rightarrow$ )
- 2 if and only if ( $\leftrightarrow$ )



# Connective: not

$p$ : a proposition.

**Negation of  $p$ :**  $\neg p$  or  $\bar{p}$  “It is not the case that  $p$ .”

**Example:**

Find the negation of “Michael’s PC runs Linux”

**Solution:**

“It is not the case that Michael’s PC runs Linux.”

**Alternative expressions:**

A proposition: Einstein is a genius

Negation:

It is not the case that Einstein is a genius,

or

It is false that Einstein is a genius,

or

Einstein is not a genius.

**Truth table:**

| $p$ | $\neg p$ |
|-----|----------|
| $T$ | $F$      |
| $F$ | $T$      |

# Connective: and/conjunction

Let  $p$  and  $q$  be propositions. The conjunction of  $p$  and  $q$ , denoted by  $p \wedge q$ , is the proposition “ $p$  and  $q$ ”. The conjunction  $p \wedge q$  is true when both  $p$  and  $q$  are true and is false otherwise.

| $p$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| $T$ | $T$ | $T$          |
| $T$ | $F$ | $F$          |
| $F$ | $T$ | $F$          |
| $F$ | $F$ | $F$          |

**Table:** The Truth Table for the Conjunction of Two Propositions.

# Connective: or/disjunction

Let  $p$  and  $q$  be propositions. The disjunction of  $p$  and  $q$ , denoted by  $p \vee q$ , is the proposition “ $p$  or  $q$ ”. The disjunction  $p \vee q$  is false when both  $p$  and  $q$  are false and is true otherwise.

| $p$ | $q$ | $p \vee q$ |
|-----|-----|------------|
| $T$ | $T$ | $T$        |
| $T$ | $F$ | $T$        |
| $F$ | $T$ | $T$        |
| $F$ | $F$ | $F$        |

**Table:** The Truth Table for the Disjunction of Two Propositions.

# Connective: implication/if ... then

$p, q$ : propositions.

The conditional statement  $p \rightarrow q$  “if  $p$ , then  $q$ ”.

The conditional statement  $p \rightarrow q$  is false when  $p$  is true and  $q$  is false, and true otherwise.

$p$ : the hypothesis (or antecedent or premise)

$q$ : the conclusion (or consequence).

## Ways to express $p \rightarrow q$ :

“if  $p$ , then  $q$ ”

“ $p$  is sufficient for  $q$ ”

“ $q$  when  $p$ ”

“ $q$  unless  $\neg p$ ”

“ $p$  only if  $q$ ”

“ $q$  whenever  $p$ ”

“ $q$  follows from  $p$ ”

“if  $p, q$ ”

“ $q$  if  $p$ ”

“a necessary condition for  $p$  is  $q$ ”

“ $p$  implies  $q$ ”

“a sufficient condition for  $q$  is  $p$ ”

“ $q$  is necessary for  $p$ ”

# Connective: implication/if ... then

| $p$ | $q$ | $p \rightarrow q$ |
|-----|-----|-------------------|
| $T$ | $T$ | $T$               |
| $T$ | $F$ | $F$               |
| $F$ | $T$ | $T$               |
| $F$ | $F$ | $T$               |

**Table:** The Truth Table for the the Conditional Statement  $p \rightarrow q$ .

**Example:** If there is lockdown, then you cannot go outside.

$p$ : there is lockdown

$q$ : you can not go outside

If there is a lockdown, then you cannot go outside.

If there is no lockdown, you may or may not go outside depending on other factors.

# Converse, Contrapositive, Inverse

$p \rightarrow q$

**Converse:**  $q \rightarrow p$

**Contrapositive:**  $\neg q \rightarrow \neg p$

**Inverse:**  $\neg p \rightarrow \neg q$

**Example:** “The home team wins whenever it is raining?”

$p$ : it is raining

$q$  the home team wins

$p \rightarrow q$ : “If it is raining, then the home team wins”

**Converse:** “If the home team wins, then it is raining”

**Contrapositive:** “If the home team does not win, then it is not raining”

**Inverse:** “If it is not raining, then the home team does not win”

**Remark:** Contrapositive is equivalent to the original statement.

# Biconditionals/if and only if

$p, q$ : propositions.

**Biconditional statement:**  $p \leftrightarrow q$  “ $p$  if and only if  $q$ ”.

$p \leftrightarrow q$  is true when  $p$  and  $q$  have the same truth values and is false otherwise.

**Ways to express  $p \leftrightarrow q$ :**

“ $p$  is necessary and sufficient for  $q$ ”

“if  $p$  then  $q$ , and conversely”

“ $p$  iff  $q$ ”

# Biconditionals/if and only if

| $p$ | $q$ | $p \leftrightarrow q$ |
|-----|-----|-----------------------|
| $T$ | $T$ | $T$                   |
| $T$ | $F$ | $F$                   |
| $F$ | $T$ | $F$                   |
| $F$ | $F$ | $T$                   |

**Table:** The Truth Table for the the Conditional Statement  $p \rightarrow q$ .



# Compound Propositions

- Complicated compound propositions are formed using connectives.
- It involves any number of propositional variables.
- Truth tables can be used to determine the truth values of these propositions.

**Example:**  $((p \wedge q) \vee \neg r) \leftrightarrow p$

| $p$ | $q$ | $r$ | $(p \wedge q)$ | $\neg r$ | $((p \wedge q) \vee \neg r)$ | $((p \wedge q) \vee \neg r) \leftrightarrow p$ |
|-----|-----|-----|----------------|----------|------------------------------|------------------------------------------------|
| T   | T   | T   | T              | F        | T                            | T                                              |
| T   | T   | F   | T              | T        | T                            | T                                              |
| T   | F   | T   | F              | F        | F                            | F                                              |
| T   | F   | F   | F              | T        | T                            | T                                              |
| F   | T   | T   | F              | F        | F                            | T                                              |
| F   | T   | F   | F              | T        | T                            | F                                              |
| F   | F   | T   | F              | F        | F                            | T                                              |
| F   | F   | F   | F              | T        | T                            | F                                              |

**Table:** Truth Table for the the Compound Statement  $((p \wedge q) \vee \neg r) \leftrightarrow p$ .

# Precedence of Logical Operators

| Operator          | Precedence |
|-------------------|------------|
| $\neg$            | 1          |
| $\wedge$          | 2          |
| $\vee$            | 3          |
| $\rightarrow$     | 4          |
| $\leftrightarrow$ | 5          |

Table: Precedence of Logical Operators

**Example:**  $p \wedge q \vee r$  means  $(p \wedge q) \vee r$

# Propositional Equivalences

**Tautology:** A compound proposition that is always true.

**Example:**  $p \vee \neg p$

**Contradiction:** A compound proposition that is always false.

**Example:**  $p \wedge \neg p$

**Contingency:** A compound proposition that is neither a tautology nor a contradiction.

# Logical Equivalences

**Logical Equivalences:**  $P$  and  $Q$  are equivalent if they have the same truth tables.

$P \equiv Q$  to mean  $P$  is equivalent to  $Q$ .

The compound propositions  $P$  and  $Q$  are called logically equivalent if  $P \leftrightarrow Q$  is a tautology.

One way to determine whether two compound propositions are equivalent is to use a truth table.

# De Morgan's laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

| $p$ | $q$ | $(p \wedge q)$ | $\neg(p \wedge q)$ | $\neg p$ | $\neg q$ | $\neg p \vee \neg q$ |
|-----|-----|----------------|--------------------|----------|----------|----------------------|
| T   | T   | T              | F                  | F        | F        | F                    |
| T   | F   | F              | T                  | F        | T        | T                    |
| F   | T   | F              | T                  | T        | F        | T                    |
| F   | F   | F              | T                  | T        | T        | T                    |

Show that  $p \rightarrow q$  and  $\neg p \vee q$  are logically equivalent.

| $p$ | $q$ | $\neg p$ | $\neg p \vee q$ | $p \rightarrow q$ |
|-----|-----|----------|-----------------|-------------------|
| T   | T   | F        | T               | T                 |
| T   | F   | F        | F               | F                 |
| F   | T   | T        | T               | T                 |
| F   | F   | T        | T               | T                 |

# Logical Equivalences

| Equivalence                                                                                                              | Name                |
|--------------------------------------------------------------------------------------------------------------------------|---------------------|
| $p \wedge T \equiv p, \quad p \vee F \equiv p$                                                                           | Identity laws       |
| $p \vee T \equiv T, \quad p \wedge F \equiv F$                                                                           | Domination laws     |
| $p \vee p \equiv p, \quad p \wedge p \equiv p$                                                                           | Idempotent laws     |
| $\neg(\neg p) \equiv p$                                                                                                  | Double negation law |
| $p \vee q \equiv q \vee p, \quad p \wedge q \equiv q \wedge p$                                                           | Commutative laws    |
| $(p \vee q) \vee r \equiv p \vee (q \vee r)$<br>$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$                     | Associative laws    |
| $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$<br>$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | Distributive laws   |
| $\neg(p \wedge q) \equiv \neg p \vee \neg q$<br>$\neg(p \vee q) \equiv \neg p \wedge \neg q$                             | De Morgan's laws    |
| $p \vee (p \wedge q) \equiv p, \quad p \wedge (p \vee q) \equiv p$                                                       | Absorption laws     |
| $p \vee \neg p \equiv T, \quad p \wedge \neg p \equiv F$                                                                 | Negation laws       |
| $(p \rightarrow q) \equiv \neg p \vee q$                                                                                 | implication         |
| $(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$                                                                   | contrapositive      |

# Converse, Contrapositive, Inverse

**Proposition:**  $p \rightarrow q$

**Converse:**  $q \rightarrow p$

**Contrapositive:**  $\neg q \rightarrow \neg p$

**Inverse:**  $\neg p \rightarrow \neg q$

The truth table:

| $p$ | $q$ | $\neg p$ | $\neg q$ | $p \rightarrow q$ | $q \rightarrow p$ | $\neg q \rightarrow \neg p$ | $\neg p \rightarrow \neg q$ |
|-----|-----|----------|----------|-------------------|-------------------|-----------------------------|-----------------------------|
| T   | T   | F        | F        | T                 | T                 | T                           | T                           |
| T   | F   | F        | T        | F                 | T                 | F                           | T                           |
| F   | T   | T        | F        | T                 | F                 | T                           | F                           |
| F   | F   | T        | T        | T                 | T                 | T                           | T                           |

# Constructing Logical Equivalences without truth tables

**Example:** Show that  $\neg(p \rightarrow q)$  and  $p \wedge \neg q$  are logically equivalent.

$$\begin{aligned}\neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && \text{by } p \rightarrow q \equiv \neg p \vee q \\ &\equiv \neg(\neg p) \wedge \neg q && \text{by the second De Morgan law} \\ &\equiv p \wedge \neg q && \text{by the double negation law}\end{aligned}$$

**Example:** Show that  $\neg(p \vee (\neg p \wedge q))$  and  $\neg p \wedge \neg q$  are logically equivalent.

$$\begin{aligned}\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by second De Morgan law} \\ &\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by first De Morgan law} \\ &\equiv \neg p \wedge (p \vee \neg q) && \text{by the double negation law} \\ &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the second distributive law} \\ &\equiv F \vee (\neg p \wedge \neg q) && \text{because } \neg p \wedge p \equiv F \\ &\equiv (\neg p \wedge \neg q) \vee F && \text{by commutative law for disjunction} \\ &\equiv \neg p \wedge \neg q && \text{by the identity law for } F\end{aligned}$$



# Logical Inferences

**Expression:**  $\phi_1, \phi_2, \dots, \phi_n \vdash \psi$

- This expression is called **sequent**.
- $\phi_1, \phi_2, \dots, \phi_n$  are called **premises**.
- $\psi$  is called **conclusion**.
- Given premises one has to reach a conclusion.

**Example:** If you have money then you can buy food. You have money. Therefore you can buy food.

**Answer:**

$p$ : you have money

$q$ : you can buy food

$p \rightarrow q$

$p$

---

$\therefore q$

# Logical Inferences

Inferences can be written as:

- (conjunction of premises)  $\rightarrow$  (conclusion).  
**Valid inferences:** (conjunction of premises)  $\rightarrow$  (conclusion) is a tautology
- All premises are true
- **Invalid inferences:** Not tautology

# Inference rules

| Rule of Inference                                                                | Tautology                                                                    | Name                   |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------|
| $\frac{p}{\therefore (p \vee q)}$                                                | $p \rightarrow (p \vee q)$                                                   | Addition               |
| $\frac{(p \wedge q)}{\therefore p}$                                              | $(p \wedge q) \rightarrow p$                                                 | simplification         |
| $\frac{p \quad p \rightarrow q}{\therefore q}$                                   | $(p \wedge (p \rightarrow q)) \rightarrow q$                                 | Modus ponens           |
| $\frac{\neg q \quad p \rightarrow q}{\therefore \neg p}$                         | $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$                       | Modus tollens          |
| $\frac{(p \rightarrow q) \quad (q \rightarrow r)}{\therefore (p \rightarrow r)}$ | $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ | Hypothetical syllogism |

# Inference rules

| Rule of Inference                                                | Tautology                                                    | Name                  |
|------------------------------------------------------------------|--------------------------------------------------------------|-----------------------|
| $\frac{(p \vee q) \quad \neg p}{\therefore q}$                   | $((p \vee q) \wedge \neg p) \rightarrow q$                   | Disjunctive syllogism |
| $\frac{(p) \quad (q)}{\therefore (p \wedge q)}$                  | $((p) \wedge (q)) \rightarrow (p \wedge q)$                  | Conjunction           |
| $\frac{(p \vee q) \quad (\neg p \vee r)}{\therefore (q \vee r)}$ | $((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$ | Resolution            |

# Build Arguments: using inference

Show the premises “It is not sunny this afternoon and it is colder than yesterday,” “We will go swimming only if it is sunny,” “If we do not go swimming, then we will take a canoe trip,” “If we take a canoe trip, then we will be home by sunset” lead to the conclusion “We will be home by sunset.”

## Answer:

$p$  “It is sunny this afternoon,”       $q$  “It is colder than yesterday”  
 $r$  “We will go swimming”       $s$  “We will take a canoe trip”  
 $t$  “We will be home by sunset.”

Premises  $\neg p \wedge q$ ,  $r \rightarrow p$ ,  $\neg r \rightarrow s$ , and  $s \rightarrow t$ .

Conclusion  $t$ .

| Step                      | Reason                          |
|---------------------------|---------------------------------|
| 1. $\neg p \wedge q$      | Premise                         |
| 2. $\neg p$               | Simplification using (1)        |
| 3. $r \rightarrow p$      | Premise                         |
| 4. $\neg r$               | Modus tollens using (2) and (3) |
| 5. $\neg r \rightarrow s$ | Premise                         |
| 6. $s$                    | Modus ponens using (4) and (5)  |
| 7. $s \rightarrow t$      | Premise                         |
| 8. $t$                    | Modus ponens using (6) and (7)  |

# Predicate Logic

# Introduction

Everyone who study hard get a good grade

John studied hard

∴ John get a good grade

Raj does not get a good grade

Predicate logic  $\rightarrow$  Proposition logic + something else

# Predicate Logic

Consider the statements

$$\bullet \quad x > 3 \qquad \qquad x = y + 3 \qquad \qquad x + y = z$$

$>, +, =$ : relations

- The truth value of these statements is **not** valid without specifying the value of  $x, y, z$ .
- The proposition can be created by specifying values for the variables.
- A predicate is a property that is **affirmed** (*true*) or **denied** (not *true*) about the subject (variable or argument).

**Example:**  $x$  is greater than 3.

$x$ : subject and the rest is predicate.



# Propositional function

$P(x)$  :  $x$  is greater than 3.

- $P(x)$  is a statement
- $P$  is a predicate or propositional function
- $x$  is an argument

## Example:

- $Boy(x)$ :  $x$  is a boy (Unary predicate)
- $Brother(x, y)$ :  $x$  is a brother of  $y$  (Binary predicate)
- $Sum(x, y, z)$ : Is  $x + y = z$ ? (Ternary predicate)
- $P(x_1, x_2, x_3, \dots, x_n)$ :  $n$ -ary predicate

# Propositional function

- Consider a propositional function  $P(x_1, x_2, x_3, \dots, x_n)$
- This is a function that
  - evaluates to *true* or *false*.
  - takes one or more arguments.
  - express a predicate involving the arguments
  - becomes a proposition when the values of the arguments are assigned.

**Example:**  $P(x, y, z): x^2 + y^2 = z^2$

- The truth value of  $P(3, 4, 5)$ : *true*
- The truth value of  $P(4, 5, 7)$ : *false*
- Number of such assignments: infinitely many

# Universe of discourse

- Consider the statement  $x > 3$   
Assign the values of  $x$  as Apple, or blue, or John  
It makes no sense.
- **Universe of discourse:** The set of all objects that we can sensibly assign to a variable in a propositional function.

**Universe of discourse, multivariate function:** Consider the function  $P(x_1, x_2, \dots, x_n)$

- For each variable, the universe of discourse can be different.

**Example:**  $P(x, y, z)$  : The sum of an integer and a real number is a real  
 $P(2, 2.5, 4.5)$  is *true*  
 $P(3, 5.5, 4.5)$  is *false*  
 $P(5.1.5, 4.5)$  does not make sense, as 5.1 is not an integer

# Quantifiers

- Recall the statement:  $x > 3$
- This is not a proposition
- It becomes a proposition when we assign values to  $x$ 
  - $10 > 3$  *true*
  - $2 > 3$  *false*
- Another way: by quantifying the statement.

## Quantifiers:

- Universal quantifiers ( $\forall$ )
  - The proposition is true for **all** possible values in the universe of discourse
- Existential quantifiers ( $\exists$ )
  - The proposition is true for **some** values in the universe of discourse

# Universal quantifiers

The universal quantification of a predicate  $P(x)$  is a proposition  
“ $P(x)$  is true for all values in the universe of discourse”

Denoted by  $\forall x P(x)$

- If the universe of discourse is  $\{n_1, n_2, \dots, n_k\}$  then,  
$$\forall x P(x) = P(n_1) \wedge P(n_2) \wedge \dots \wedge P(n_k)$$

**Example:**  $P(x)$ :  $x$  is a first year student.

$Q(x)$ :  $x$  must take discrete mathematics course.

**Express:** Every first year student must take discrete mathematics course

$$\forall x (P(x) \rightarrow Q(x))$$

Every student must take discrete mathematics course or they are not a first year student

$$\forall x (Q(x) \vee \neg P(x))$$

For every  $x$  and every  $y$ ,  $x + y > 10$

$P(x, y)$ :  $x + y > 10$

$$\forall x \forall y P(x, y)$$

# Existential quantifiers

The existential quantification of a predicate  $P(x)$  is a proposition  
“ $P(x)$  is true for some values in the universe of discourse”

Denoted by  $\exists x P(x)$

- If the universe of discourse is  $\{n_1, n_2, \dots, n_k\}$  then,  
 $\exists x P(x) = P(n_1) \vee P(n_2) \vee \dots \vee P(n_k)$

**Example:** There is a value of  $x$  that satisfy  $x + 7 = 10$

$P(x): x + 7 = 10$

$\exists x P(x)$

# Existential quantifiers: example

**Example:** Express the statement  
“there exists a real solution to  $ax^2 + bx + c = 0, a \neq 0$ ”

- $P(x): x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- universe of discourse for  $x$  is real numbers.
- the statement can be represented as  $\exists x P(x)$ .
- truth value of  $\exists x P(x)$ 
  - It is true when  $b^2 - 4ac \geq 0$
  - It is false when  $b^2 - 4ac < 0$

# Truth values of quantifiers

|                  | When true                                   | when false                                   |
|------------------|---------------------------------------------|----------------------------------------------|
| $\forall x P(x)$ | for every $x$ , $P(x)$ is true              | there exists a $x$ for which $P(x)$ is false |
| $\exists x P(x)$ | there exists a $x$ for which $P(x)$ is true | for every $x$ , $P(x)$ is false              |

- $\forall$  and  $\exists$  both can be present in a statement, i.e.,  $\forall x \exists y P(x, y)$

## Remark:

- The quantifiers must be read from left to right.
- The order is important:  $\forall x \exists y P(x, y) \neq \exists y \forall x P(x, y)$
- $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$   
 $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$



# Examples

**Example:** Express in predicate logic  
“there is an infinite number of integers”

**Answer:**

- $P(x, y): x < y$
- The universe of discourse is integer
- The expression is  $\forall x \exists y P(x, y)$

**Example:** Law of addition:  
“for every pair of reals the following holds,  $x + y = y + x$ ”

**Answer:**

- $P(x, y): x + y$
- The universe of discourse is real numbers
- The expression is  $\forall x \forall y (P(x, y) \leftrightarrow P(y, x))$

# Binding variables

**Bound:** When a quantifier is used on a variable then the variable is bound.

**Free:** If no quantifier is used on a variable then the variable is free.

**Example:**  $\exists x \forall y P(x, y)$  : Both  $x$  and  $y$  are bound

$\forall x \exists z P(x, y, z)$  :  $x, z$  are bound,  $y$  is free

**Well-formed formula:** All variables are bound or all variables are properly quantified.

**Example:**  $\exists x, y \forall z P(x, y, z, w)$

- Scope of existential quantifier  $x, y$
- Scope of universal quantifier  $z$
- free variable  $w$
- It is not well-formed

# Negation

Let  $P(x)$  be a predicate

- $\neg(\forall x P(x)) \equiv \exists x P(x)$
- $\neg(\exists x P(x)) \equiv \forall x P(x)$

# Introduction to Proofs

# Introduction to Proofs

To establish that something holds.

Why is it important?

- Validate some statement
- An algorithm needs to be proved correct.
- Prove some properties of data structure.
- ...

## Not valid proof techniques:

- Proof by example
- Proof by simulation
- Proof by fame

# Some terminologies

**Theorem:** A theorem is a statement that can be shown to be true (by providing a proof).

**Proof:** A sequence of statements that forms an argument.

**Axioms/postulates:** Statements that are assumed to be true

- Two types of theorems.

**Lemma:** A less important theorem that is used to prove some other result.

**Corollary:** A theorem that can be established directly from a theorem that has been proved.

**Conjecture:** Statement that is being proposed to be a true statement

# Methods of Proving Theorems

- Trivial proofs
- Vacuous proofs
- Direct proofs
- Proof by contrapositive
- Proof by contradiction
- Proof by cases
- Proof by equivalence (iff)
- Proof by counterexample

# Trivial proofs

- Conclusion is arrived without using the premises
- Show that the conclusion is always true
- show that from  $p \rightarrow q$ ,  $q$  is always true without  $p$

**Example:** Prove that “if  $x > 0$ , then  $(x + 1)^2 - 2x \geq x^2$ ”.

**Solution:**  $(x + 1)^2 - 2x = x^2 + 2x + 1 - 2x = x^2 + 1 \geq x^2$



# Vacuous proofs

- If the premise  $p$  is false then  $p \rightarrow q$  is always true
- It is a proof that is based on the fact that no element in the universe of discourse satisfies the premise

**Example:** If  $x$  is a prime number divisible by 6 then  $x^2 < 0$ .

# Direct proofs

- It is a common type of proof.
- Assume hypothesis  $p$
- Provides a direct series of implications.
- Use the rules of inferences
- It can use some other results (proved independently).
- Show that the conclusion is true.

**Example:** Prove that  $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ .

**Solution:**  $x = 1 + 2 + 3 + \dots + n$

$$x = n + (n - 1) + (n - 2) + \dots + 1$$

$$2x = (n + 1) + (n + 1) + (n + 1) + \dots + (n + 1)$$

$$= n(n + 1), \text{ so, } x = \frac{n(n+1)}{2}$$

# Proof by contrapositive

- Given statement  $p \rightarrow q$ , contrapositive statement  $\neg q \rightarrow \neg p$
- Assume that the conclusion is false.
- Then give a series of implications to show that the premise is false.

**Example:** Prove that “if  $x^3 < 0$ , then  $x < 0$ ”.

**Solution:**  $p : x^3 < 0$ ,  $\neg p : x^3 \geq 0$

$q : x < 0$ ,  $\neg q : x \geq 0$

Need to prove:  $\neg q \rightarrow \neg p$ , i.e., if  $x \geq 0$  then  $x^3 \geq 0$

Now,  $x^3 = x \cdot x \cdot x \geq 0$

# Proof by contradiction

- Need to prove that  $p$  is true
- Assume a contradiction  $q$  is such that  $\neg p \rightarrow q$  is true.
- $q$  is false and  $\neg p \rightarrow q$  is true, then  $\neg p$  is false.
- $p$  is true

How do we obtain a contradiction?

Suppose that  $r$  is a statement, then  $(r \wedge \neg r)$  is a contradiction.

take  $q$  as  $(r \wedge \neg r)$

For a known result  $r$ , show that  $\neg p \rightarrow (r \wedge \neg r)$  is true.

**Example:** At least 4 of any 22 days must fall on the same day of the week.

**Solution:**  $p$  : At least 4 of any 22 days must fall on the same day of the week.

Assume that  $\neg p$  is true i.e., at most 3 of any 22 days must fall on the same day of the week.

there are 7 different days in a week each day have at most 3 occurrences  
there are at most 21 days could have been chosen.

$r$ : 22 days are chosen

So  $\neg r$  is also true

Therefore,  $\neg p \rightarrow (r \wedge \neg r)$

$(r \wedge \neg r)$  is a contradiction, hence  $\neg p$  is false and then  $p$  is true

# Proof by contradiction: proving conditional statements

$p \rightarrow q \equiv (p \wedge \neg q) \rightarrow F$  (contradiction) can be proved using truth table

Assume the negation of the conclusion then the premise and the negation of the conclusion lead to a contradiction.

$p$  and  $\neg q$  are true. We know that  $\neg q \rightarrow \neg p$  (contrapositive)

this leads  $\neg p$  is true

a contradiction  $p \wedge \neg p$

**Example:** If  $3n + 2$  is odd then  $n$  is odd.

**Solution:**  $p : 3n + 2$  is odd

$q : n$  is odd

Assume  $p$  and  $\neg q$  are true.

$3n + 2$  is odd and  $n$  is not odd i.e.,  $n$  is even

$$n = 2k$$

$$3n + 2 = 6k + 2 = 2(3k + 1)$$

So  $3n + 2$  is even

$3n + 2$  is not odd so  $\neq p$

So  $p \wedge \neg p$

# Proof by cases

Break a statement into different cases and prove each of them separately.

**Example:** Prove that for an integer  $n$ ,  $9n^2 = 3n - 2$  is even.

**Solution:**  $9n^2 = 3n - 2 = 8n^2 + n^2 + 2n + n - 2 = 8n^2 + 2n - 2 + n^2 + n$   
 $= 2(4n^2 + n - 1) + n(n + 1)$

Case 1:  $2(4n^2 + n - 1)$  is even

Case 2:  $n(n + 1)$  is even

Another way to prove:

Case 1: Assume that  $n$  is even i.e.,  $n = 2k$

$$9n^2 = 3n - 2 = 9(2k)^2 + 3(2k) - 2 = 9 \cdot 4k^2 + 6k - 2 = 2(18k^2 + 3k - 1)$$

Case 2: Assume that  $n$  is odd i.e.,  $n = 2k + 1$

$$9n^2 = 3n - 2 = 9(2k + 1)^2 + 3(2k + 1) - 2 = 9(4k^2 + 2k + 1) + (6k + 3) - 2 =$$
$$(36k^2 + 36k + 9 + 6k + 3 - 2) = 36k^2 + 42k + 10 = 2(18k^2 + 21k + 5)$$

# Proof by equivalence (if and only if)

- To show an equivalence  $p \leftrightarrow q$
- $p \leftrightarrow q \equiv p \rightarrow q \wedge q \rightarrow p$

**Example:** Prove that “ $x$  is odd iff  $x^2 + 2x + 1$  is even”.

**Solution:** Assume that  $x$  is odd i.e.,  $x = 2k + 1$  for some  $k$

Now,  $x^2 + 2x + 1 = (x + 1)^2 = (2k + 1 + 1)^2 = (2k + 2)^2 = 4(k + 1)^2$

Assume that  $x^2 + 2x + 1$  is even

$$x^2 + 2x + 1 = 4k^2 \Rightarrow (x + 1) = 2k \Rightarrow x = 2k - 1$$

# Proof by equivalence (if and only if)

Let a theorem have many propositions, and the propositions are equivalent.

$p_1, p_2, \dots, p_n$  can be written as,  $p_1 \leftrightarrow p_2 \leftrightarrow p_3 \leftrightarrow \dots \leftrightarrow p_n$

To prove this statement, one needs to show

$p_1 \rightarrow p_2 \rightarrow p_3 \rightarrow \dots \rightarrow p_n \rightarrow p_1$

Or equivalently,  $p_1 \rightarrow p_2, p_2 \rightarrow p_3, \dots, p_{n-1} \rightarrow p_n, p_n \rightarrow p_1$

**Example:** Prove that the following statements are equivalent,  $n$  is even,  $n - 1$  is odd,  $n^2$  is even.

**Solution:**  $p_1$ :  $n$  is even,  $p_2$ :  $n - 1$  is odd, and  $p_3$ :  $n^2$  is even.  
We need to show that ,  $p_1 \rightarrow p_2, p_2 \rightarrow p_3, p_3 \rightarrow p_1$



# Proof by counterexample

- To show some statement is not true

**Example:** Disprove that “ $n^2 + n + 1$  is a prime number for  $n \geq 1$ ”.

**Solution:**  $n = 1 : n^2 + n + 1 = 1$

$n = 2 : n^2 + n + 1 = 7$

$n = 3 : n^2 + n + 1 = 13$

$n = 4 : n^2 + n + 1 = 21$  Not prime